Technical Document #430: Retrieval Augmented Generation - Comprehensive Overvi

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Query decomposition breaks complex queries into simpler sub-queries.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.